

SQL for Data Analytics — Retail Sales Exploration

Section A: Concept Application

1. What is the functional difference between `SELECT *` and specifying column names, and when is each preferred?

`SELECT *` fetches all columns from the table, while specifying column names (e.g., `SELECT OrderID, Sales, Profit`) fetches only the ones listed. `SELECT *` is preferred for quick, ad-hoc exploration when the structure is unfamiliar or a full view is needed. Specifying columns is preferred in production queries and reports, since it reduces data transfer, improves readability, and avoids breaking downstream code if the table structure changes later.

2. Which keyword renames a column in the output, and does this alias change the actual table structure in the database?

The `AS` keyword renames a column (creates an alias) in the query output, e.g., `SELECT Profit AS Earnings`. This alias only affects how the column is displayed in that query's result set — it does not change the actual table structure, column name, or schema stored in the database.

3. Why does wrapping a numeric value in quotes (e.g., `'5000'`) in a `WHERE` clause create a data type conflict in SQL?

Writing `WHERE Sales = '5000'` tells SQL to treat 5000 as a string/text value rather than a number. If the `Sales` column is numeric (`INT`, `FLOAT`, `DECIMAL`), SQL either throws a data type mismatch error or performs an implicit conversion, which can cause unexpected behavior, prevent index usage, and hurt performance. Numeric literals should be written without quotes; quotes are reserved for strings and dates.

4. Contrast the results of `ORDER BY Profit DESC` versus `ASC` when the goal is to identify the top 10 most profitable orders.

`ORDER BY Profit DESC` sorts from highest to lowest, bringing the most profitable orders to the top — this is what should be used to identify the top 10 most profitable orders. `ORDER BY Profit ASC` sorts from lowest to highest, surfacing the least profitable or most loss-making orders first, which is the opposite of the intended goal.

5. What is the T-SQL equivalent of the `LIMIT` clause in MS SQL Server, and why does syntax vary across SQL engines?

MS SQL Server does not use `LIMIT`; instead it uses `TOP`, e.g., `SELECT TOP 10 * FROM Orders ORDER BY Profit DESC`. Syntax varies across engines because each vendor (MySQL, PostgreSQL, SQL Server, Oracle) developed its own SQL dialect on top of the ANSI SQL standard, and row-limiting was never fully standardized — MySQL/PostgreSQL use `LIMIT`, SQL Server uses `TOP`, and Oracle traditionally used `ROWNUM` (with newer versions supporting `FETCH FIRST n ROWS ONLY`).

6. Explain the logical execution order of a query containing `SELECT`, `WHERE`, `ORDER BY`, and `LIMIT` clauses.

Although a query is written as `SELECT`, `FROM`, `WHERE`, `ORDER BY`, `LIMIT`, SQL executes it in a different logical order: (1) `FROM` identifies the data source, (2) `WHERE` filters rows based on conditions, (3) `SELECT` chooses the columns/expressions to return, (4) `ORDER BY` sorts the resulting rows, and (5) `LIMIT` restricts the final output to a set number of rows. This is why a `SELECT`-defined alias generally cannot be used inside `WHERE`, but can be used in `ORDER BY`.

Section B: Practical Task

1. Execute a query to retrieve the first 20 records from the orders table to verify data ingestion.

```
SELECT TOP 20 *  
FROM orders;
```

```
-- MySQL / PostgreSQL equivalent:  
SELECT * FROM orders LIMIT 20;
```

2. Select Order ID, Order Date, Sales, and Profit, applying a column alias to display Sales as Total_Sales.

```
SELECT  
    [Order ID],  
    [Order Date],  
    Sales AS Total_Sales,  
    Profit  
FROM orders;
```

3. Filter the dataset to isolate all high-value transactions where the Sales figure exceeds 5000.

```
SELECT *  
FROM orders  
WHERE Sales > 5000;
```

4. Generate a report of the top 10 most profitable orders by sorting the records by Profit in descending order.

```
SELECT TOP 10 *  
FROM orders  
ORDER BY Profit DESC;
```

```
-- MySQL / PostgreSQL equivalent:  
SELECT * FROM orders ORDER BY Profit DESC LIMIT 10;
```